# Cunha Lab · Invertebrate Specimen Database

## Administrator Guide

*Supabase backend + web application*

## 1. System Overview

The database consists of two parts: a Supabase backend (PostgreSQL database + authentication) hosted at supabase.com, and a web application hosted at cunhalabdb.netlify.app. As admin you might manage both.

| Component | Purpose |
|---|---|
| `Supabase` | Database, authentication, row-level security, data storage |
| `Netlify app` | Web interface for viewing, searching, and editing records |

## 2. Supabase Access

Log in at supabase.com with your admin credentials. Your project is named cunha-invertebrate-specimens.

### 2.1 Key areas in Supabase

- Table Editor — browse and manually edit any table
- Authentication → Users — manage all user accounts
- SQL Editor — run SQL queries and scripts
- Storage — not currently used

> *Always use the SQL Editor for bulk operations. The Table Editor is best for viewing and small manual edits.*

# 3. Database Schema

The database has 6 tables. Data flows in this order:

| Table | Description |
|---|---|
| `EventData` | Collection events. Where and when specimens were collected |
| `SpecimenData` | Specimen data. Linked to an event via event_code |
| `DNAExtractions` | DNA extractions. Linked to a specimen via lot_id |
| `GenomicLibraries` | Genomic libraries. Linked to an extraction via extraction_id |
| `Loans` | Loan specimens. Specimens borrowed from other institutions |
| `user_roles` | User access control. Maps user IDs to viewer or editor roles |

## 3.1 ID coding standard

Follow these conventions when entering new loan data:

| ID type | Format & example |
|---|---|
| `Loan ID` | LN-24-001 — LN + 2-digit year + 3-digit counter per year |

# 4. Loading Data via CSV

All bulk data entry is done by uploading CSV files directly to Supabase. Always load tables in the correct order to avoid foreign key errors.

## 4.1 Upload order

1. EventData: Must be loaded first (no dependencies)
2. SpecimenData; Requires EventData (event_code FK)
3. DNAExtractions: Requires SpecimenData (lot_id FK)
4. GenomicLibraries: Requires DNAExtractions (extraction_id FK)
5. Loans: Independent, can be loaded any time

## 4.2 How to upload a CSV

6. Go to Supabase → Table Editor → select the table
7. Click Insert in the top right
8. Select your CSV file
9. Click Import data

### 4.3 Useful SQL for data cleanup

Delete all rows from a table (before re-importing):

```
DELETE FROM "SpecimenData";
```

Check for orphaned extractions (lot_id not in SpecimenData):

```
SELECT extraction_id, lot_id FROM "DNAExtractions" d
WHERE NOT EXISTS (SELECT 1 FROM "SpecimenData" s WHERE s.lot_id = d.lot_id);
```

## 5. User Management

All users must sign in with an email and password through the web app. Access is controlled by the user_roles table; without a role, a user can log in but cannot see any data.

### 5.1 Inviting a new user

10. Go to Supabase → Authentication → Users → Add User
11. Enter their email and a temporary password
12. Share the credentials with them
13. Assign their role (see below)

> *Supabase limits email invitations on the free tier. Use Add User manually instead.*

### 5.2 Assigning a role

Run in SQL Editor after the user account exists:

```
INSERT INTO public.user_roles (user_id, role, email)
SELECT id, 'viewer', email FROM auth.users
WHERE email = 'their@email.com';
```

### 5.3 Updating a user's access

```
UPDATE public.user_roles SET role = 'editor'
WHERE email = 'their@email.com';
```

### 5.4 Removing access

```
DELETE FROM public.user_roles
WHERE email = 'their@email.com';
```

| Role | Permissions |
|---|---|
| `viewer` | Can read and search all tables, export CSV. Cannot edit. |
| `editor` | Full read/write access. Can edit records and add loans. Has access to Admin tab. |

# 6. Row Level Security (RLS)

RLS is enabled on all tables. The policies are:

- Any authenticated user can read all tables
- Only editors can insert, update, or delete records
- Users can only read their own row in user_roles

If a user can log in but sees no data, check that their role is assigned and that the read policy exists. Run:

```
SELECT * FROM public.user_roles WHERE email = 'their@email.com';
```

If the policies are missing, re-run the Set Up Roles script from the original setup on Supabase.

# 7. Web Application

The app is hosted at cunhalabdb.netlify.app. It is a single HTML file; to update it, replace the index.html file on Netlify by dragging a new version onto the Netlify dashboard.

## 7.1 Admin tab features

- User management SQL reference
- Add Loan form to add loan specimens with auto-suggested Loan ID and ISO country dropdown

## 7.2 Editing records

Editors can click any row in any table to open a detail drawer, then click Edit record to modify fields inline. Changes save immediately to Supabase.

## 7.3 Adding a loan

14. Go to Admin tab → Add Loan section → click + Add Loan
15. Fill in the form. Loan ID is auto suggested
16. Click Save loan

## 7.4 Updating the app

17. Make changes to index.html
18. Go to netlify.com → your site → Deploys
19. Drag the new index.html onto the deploy area
20. Wait ~30 seconds for the new version to go live

## 8. Backups

Supabase automatically backs up your database on paid plans. On the free tier, do manual backups regularly by exporting CSVs from each table:

21. Go to Table Editor → select table → click Export
22. Save the CSV to a safe location
23. Repeat for all 6 tables